



Setting Up Base Cloud/GPU Environment

The essential first step in Capstone implementation - provisioning compute, storage, and networking for reproducible and scalable AI workloads.

Why GPU Environments Matter

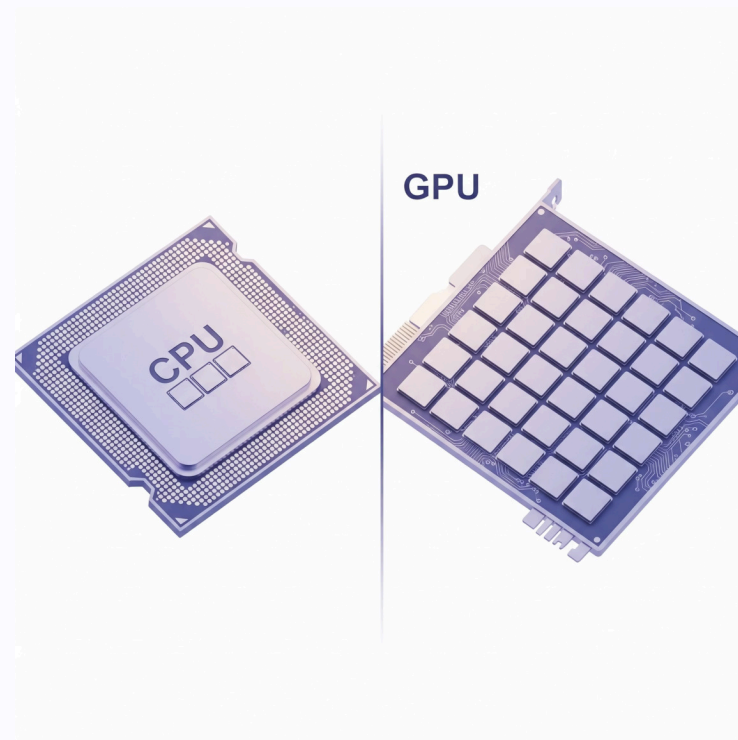
AI workloads rely heavily on linear algebra (matrix operations) that traditional CPUs struggle with.

10-100× Acceleration

GPUs and TPUs dramatically speed up neural network training compared to CPUs.

Elastic but Costly

Cloud GPUs offer flexibility but require careful right-sizing to match your [domain + metrics](#) (Day 331).



Cloud Providers for GPU Infrastructure

AWS

EC2 P4/P5 instances and SageMaker managed services with latest NVIDIA GPUs

Azure

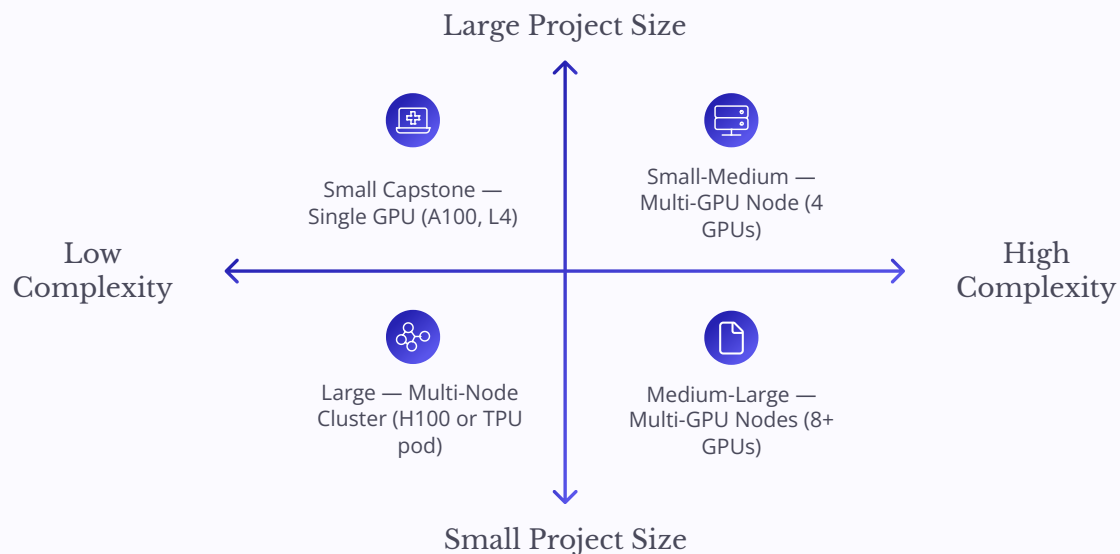
NCas, NDv4 series VMs and access to OpenAI supercomputer infrastructure

Google Cloud

A100/H100 GPU options and proprietary TPU v5e pods for specialized workloads

Your selection should be guided by [availability, budget, and compliance requirements](#) specific to your Capstone project.

Instance Selection Guide



Small Capstone

1-2 GPUs (A100, L4) - suitable for prototype models and smaller datasets

Medium Capstone

4-8 GPUs on multi-GPU node - for more complex models or larger training data

Large Capstone

TPU pod or H100 cluster - for state-of-the-art model training at scale

The key is to [match compute size to your dataset and model architecture](#).

Storage Setup

Object Storage

S3, GCS, Azure Blob - ideal for datasets and model checkpoints

Block Storage

EBS, Persistent Disk - for fast access during training cycles

Implement a [tiered approach](#):

- **Hot storage** - active training data with fastest access
- **Cold storage** - archived versions and infrequently accessed data

Always version your datasets to ensure reproducibility!



Networking Setup



VPC Isolation

Create private network segments for secure workloads



Secure Access

Implement SSH/VPN for protected remote connections



Low-Latency Links

InfiniBand/NVLink for multi-GPU communication



IAM Access

Role-based permissions for logging and monitoring



Environment Provisioning Steps

Spin up GPU instance

Launch a cloud instance with appropriate GPU specs (e.g., AWS EC2 P4d)

Attach storage

Connect EBS volumes for fast I/O and S3 buckets for datasets

Configure networking

Set up VPC, firewall rules, and appropriate IAM roles

Install drivers

Deploy CUDA/cuDNN and NVIDIA Container Toolkit

Verify GPU visibility

Confirm setup with `nvidia-smi` command

Containerized Setup (Best Practice)

Use Docker images with:

- Python + ML frameworks (PyTorch, TensorFlow, JAX)
- CUDA/cuDNN dependencies
- MLOps clients (MLflow, DVC, Weights & Biases)

✓ Key Benefit

Ensures complete **reproducibility** across development, testing, and production environments



Containerized ML environment structure

Cost Controls



Spot/Preemptible GPUs

Use discounted instances for interruptible training jobs (60-90% savings)



Auto-Shutdown Policies

Configure idle detection to automatically terminate unused resources



Utilization Tracking

Monitor GPU usage with CloudWatch, Stackdriver or custom dashboards



Budget Buffer

Always allocate 20-30% overhead for unexpected training runs

Apply lessons from [Day 333](#) to optimize your Capstone budget



Compliance & Security

Encryption

Enable at-rest and in-transit encryption for all storage buckets

Access Controls

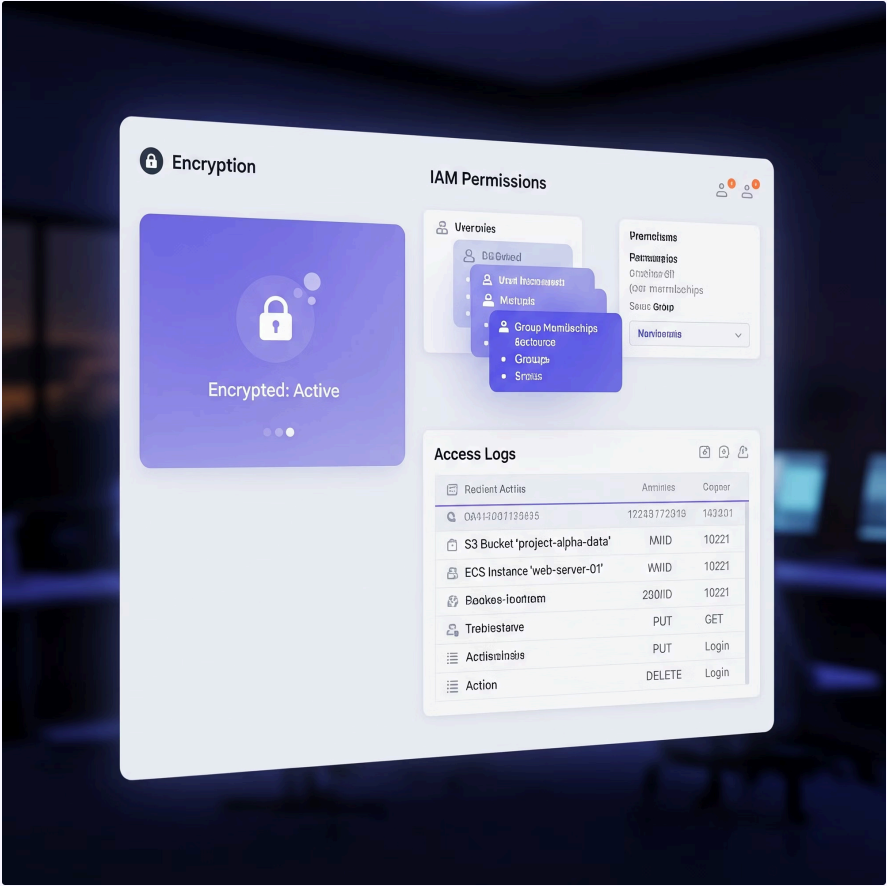
Implement IAM roles with least-privilege principle

Logging

Enable comprehensive logging with CloudTrail, Stackdriver, or Azure Monitor

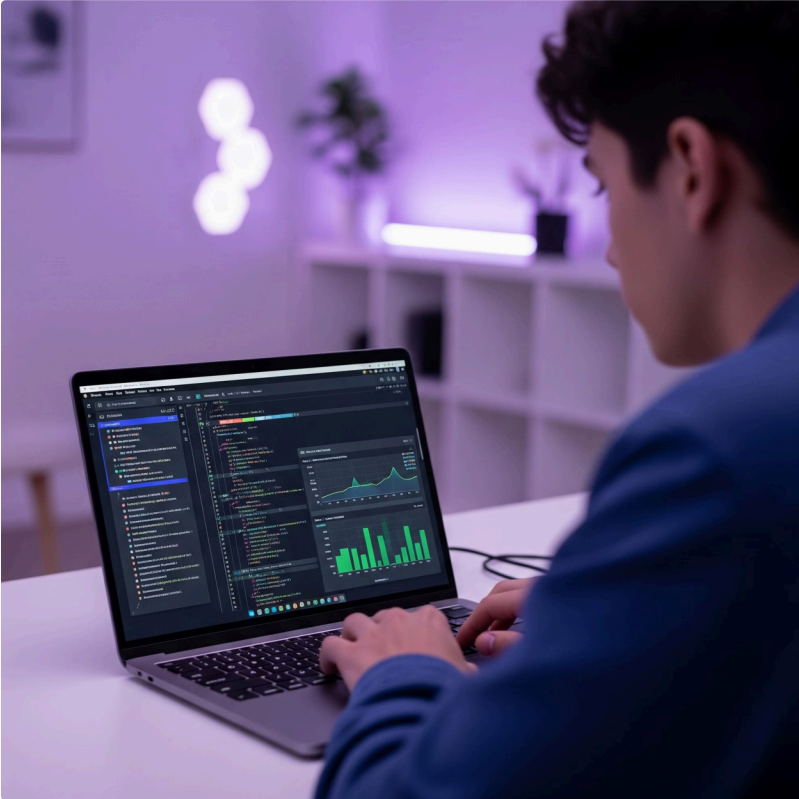
Audit

Document all access patterns for your Capstone report



Thorough security documentation demonstrates professional approach in Capstone evaluation.

Case Study: Student NLP Capstone



Implementation Details:

- Provisioned 1× A100 GPU on Google Cloud Platform
- Created Dockerized DistilBERT training environment
- Stored datasets in S3 with versioning enabled
- Tracked experiments with MLflow



Budget: Total base setup cost was under \$200, allowing room for extended training runs.

Base GPU Environment Architecture



Base GPU environment = compute + storage + networking + containers

Key Lessons

GPU environment is the foundation of your Capstone build

Reproducibility

Containerization ensures
consistent environment across all
development stages

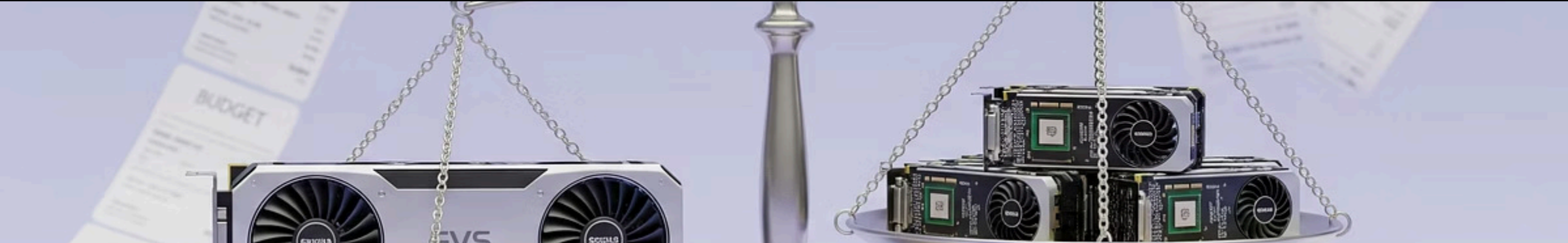
Budget Discipline

Cost and compliance
considerations must be integrated
from Day 1

Scalable Approach

Start with minimal viable
infrastructure and scale as your
project evolves

A well-designed environment lets you focus on model innovation rather than infrastructure troubleshooting.



Discussion Prompt

If you had only \$500 budget, would you set up your Capstone environment with 1 large GPU or several smaller GPUs — and why?

One Large GPU

- Better for memory-intensive models
- Simpler setup with no distributed training
- Potential for faster iterations

Several Smaller GPUs

- Enables parallel experimentation
- More resilient to single-point failures
- Good for data-parallel workloads

Consider how your specific model architecture and training approach affect this decision.